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Predicting and Categorizing Particulates in Providence: Using ML Methods to Understand PM 2.5 Regimes and Predict PM Levels with Wind Data

Abstract

PM 2.5 is a dangerous air pollutant that has a variety of sources, and is of public health concern in Providence, Rhode Island. This project aims to understand the relationship between wind conditions and the concentration of PM 2.5 throughout the city using several pollutant and wind sensors that are part of the Breathe Providence air sensor network. To achieve this, both unsupervised and supervised machine learning methods were used. A clustering technique (K-means) was used to inform on existing wind-pollutant regimes, with low to moderate success. The clusters were mainly based on wind speed, and high concentration clusters did not see unique wind directions. Thus the supervised learning method (a simple neural network) could not perform well on PM 2.5 bin prediction using only wind speed. When including other factors like PM 2.5 lags, relative humidity, and time of day, the model was able to perform better (around 75% accuracy), with permutation importance indicating wind had relatively high predictive power compared to other factors. However, permutation performance and comparison to a baseline using only the one-hour lag feature indicates that the model's performance was solely based on this one-hour lag and revealing that PM 2.5 concentration is more autocorrelated than dependent on meteorological conditions. This project serves as a proof-of-concept, and can be expanded in several ways, including using stronger machine learning methods to predict the PM 2.5 concentration directly.

Introduction

PM 2.5 refers to atmospheric particles that are under 2.5 micrometers in diameter and are produced from a variety of sources, including natural sources but are also strongly connected to industrial activity (EPA, 2025a). It is a pollutant that is of particular public health concern and is known to cause a variety of health issues, including asthma, heart problems, and other respiratory diseases (EPA, 2025b). Particulate matter is relevant to the city of Providence, which is home to a large industrial port district along the northern end of the Narragansett Bay. The port imports and exports large amounts of scrap metal, gravels, and salts, and is directly adjacent to Providence neighborhoods ("Provport"). Rhode Island is also a known hotspot for both child and adult asthma (RIDOH, 2026).

Breathe Providence (BP) was founded under these circumstances. The research group based at Brown University maintains 25 low-cost sensors (LCS) throughout the city of Providence with the goal of providing neighborhood-scale, actionable air quality data to support environmental justice and inform residents (Breathe Providence, 2026). The BP sensors are capable of measuring PM 2.5 among other pollutants. Low-cost air sensors are beneficial because of their ability to provide air pollution data at an otherwise unattainable scale due to their footprint and minimal costs (WMO, 2025).

This project serves as a proof-of-concept project for understanding the relationship between meteorological conditions and PM 2.5 concentrations in the BP network. Many projects larger projects have tackled PM 2.5 prediction, though all approaches differ slightly. Existing projects typically predict PM 2.5 at a larger scale beyond LCS networks, utilize a wider variety of meteorological conditions, and/or are based on satellite data:

- Attey-Yeboah et al. predict PM 2.5 concentrations across a LCS network in Ghana, utilizing humidity, temperature, and aerosol optical depth (AOD) data from satellites. Notably, wind is not used here.
- Cortina-Januchs et al. use both supervised and unsupervised learning methods to predict PM10 concentrations in Salamanca, Mexico. Clustering methods are used to help better understand relationships between meteorology and PM concentrations that then inform the supervised learning process. Meteorological conditions (including wind) are provided via ground-based weather stations, as opposed to satellite-based data.
- Meteorological conditions have shown to be predictive of PM 2.5 concentrations, as seen in works by Gu et al., Zamani Joharestani et al., and Zaman et al. These models use a combination of satellite based AOD data and a variety of meteorological conditions to predict PM 2.5 concentrations and are validated using ground-based sensors.
- Jin et al. and Danesh Yazdi et al. both use techniques to show that wind speed can be an important predictor of PM 2.5. Jin et al. uses a simple linear regression to show that wind is a strong predictor of PM 2.5 in specific regions of China. Danesh Yazdi et al. use more traditional XAI approach of Shapley analysis to conclude wind speed is the second most important feature for PM 2.5 prediction.

My project is twofold: K-means will be used to understand wind-pollutant regimes that exist at sites throughout Providence (representing a more informative approach), while a simple neural network will be used to predict bins of PM 2.5 concentration (as more of a predictive approach). This project allows me to work with unsupervised learning techniques as well as simple neural network architectures that I have not used in my time at BP. More details of these networks and the data used are discussed later.

This project, relative to my other work with BP, has the potential to be more informative to the Providence community. If successful, wind forecasts can be used to forecast PM 2.5 conditions and inform residents in specific neighborhoods that may be more sensitive to poor air quality. However, it should be stated the foremost goal of this work is to understand the broad connection between wind and PM 2.5 in Providence revealed through this model, not necessarily a single model output's success. Particularly this project will look at wind's predictive power in a supervised machine learning model. This project does have room for expansion. Eventually (though outside the scope of this class), other meteorological inputs could be considered alongside wind, other BP nodes could be considered that are not directly collocated with meteorological sensors, and perhaps this framework could be applied to other pollutants. More specific goals based on results of the models is discussed at the end of the paper.

Data Set

PM 2.5 data is collected from three QuantAQ Modulair sensors (Quants) which are under the BP network. These sensors are chosen over the BEACO2N sensors (which account for most of the BP network) due to their higher accuracy PM 2.5 measurements and their collocation with

wind sensors (South Coast Air Quality Management District, 2025). The three Quants are located at the following sites:

- Department of Public Works (DPW) at the corner of Allens Ave and Chapman St. This sensor is in southern Providence. This Quant does not have a Gill weather station, and thus wind data is pulled from the Allens Ave meteorological site operated by the National Oceanic and Atmospheric Administration (NOAA). The NOAA site is located 4000 feet south of the DPW Quant on the same road and obtains much of the same topography.
- Providence Emergency Management Agency (PEMA) at 591 Charles St. This site is in northern Providence
- Providence Housing Authority (PHA) at 40 Laurel Hill Ave. This site has a Gill weather station.

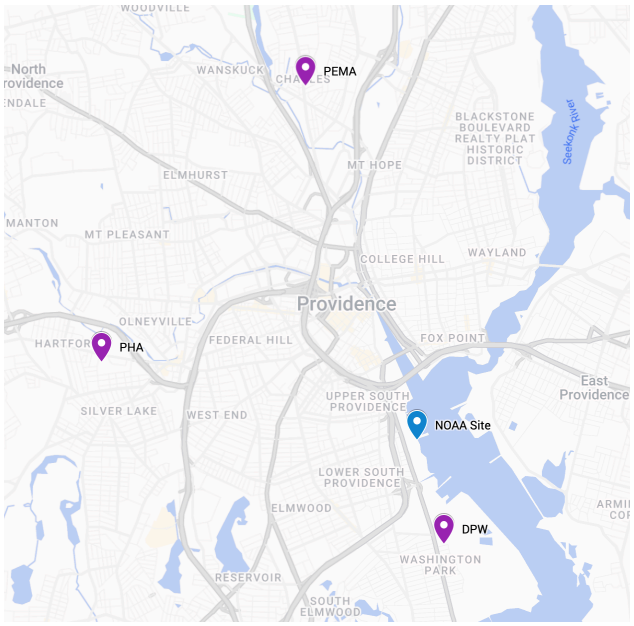


Figure 1: Sensor Location Map. These particular Quant sites are strategically chosen to be at the outer edges of Providence.

The PM 2.5 data (but not the wind data) from the Quants is visible through QuantAQ's map application (presented later in the data availability section). Live data is only downloadable as a researcher in the BP group. Wind data from Allens Ave can be accessed through the worldmet package in R ("worldmet").

The DPW site has 4938 hourly observations, PEMA has 4415 observations, and PHA has 4401 observations. Data utilized from the sites begins on January 2nd, 2025 and runs until August 24, 2025. The whole year of 2025 could not be included due to an unexplained outage of the Allens Ave site that provides meteorological data for DPW here. The difference in observations is likely due to missing values that had to be dropped (though number missing values are insignificant overall). Features used within the supervised and unsupervised learning methods are contained in table 1 in appendix.

Data preprocessing was overall minimal. Data coming from the Quants is delivered minutely and thus must be manually resampled each hour. This is done by simply averaging

minutely concentrations over the 60 observations made in an hour. Missing data points were not significant and were simply dropped. Some slightly more complex merging was needed to combine data from the NOAA Allens Ave site with the DPW data.

Some of the features above were extracted from other features. Wind data is typically fed into the Quants as polar coordinates, containing both wind speed and direction measured in degrees. To avoid continuity issues with large angle measurements (360 degrees versus 0), the u-v vector for wind was computed as inputs to the model. The hour_sin and hour_cos features were computed by taking the sine and cosine of the hour an observation was made, scaled to a 24-hour cycle. This was used as input to the supervised learning model. Lags are simply the PM 2.5 concentration taken from 1 hour before, 2 hours before, etc. at the same site. These lags ultimately proved to have strong predictive power.

Though not a feature, the pm25_bin value is produced by taking quintiles of PM 2.5 values in the time series, and labeling the lowest bin as a 0, and highest bin as a 4. The bounds for these bins at each site can be in table 2 in the appendix.

Methods

Unsupervised Methods

K-means is a useful clustering technique that is useful here for several reasons. Ideally, when running our clustering methods with our wind vectors and pollutants specific regimes may be identified that represent specific PM events/sources (i.e., a strong signal coming from a specific direction at certain wind speeds). Ideally, these clustering techniques can also provide a general understanding for how unique certain regimes are. If there are large overlaps in our clusters in terms of wind direction and speed, it may not be as informative and would be hard for an algorithm to separate these variables based on wind speed alone.

Scikit-learn's KMeans was used with the pm25, wind_u, and wind_v parameters ("scikit-learn"). Experiments were conducted at all three sites and primarily utilized 5 clusters to match the five PM 2.5 bins that will be explored in the supervised methods. All features inserted into the clustering model were scaled using z-score standardization (i.e., feature values had their mean subtracted from them, then were divided by the standard deviation). This allows for proper scaling between unique features with different units.

Additional experiments present in the code include experimenting with 9 clusters, as well as including relative humidity in the clustering techniques. Though arguably neither of these provided notable results and are thus not emphasized in the results section.

Supervised Methods

A simple neural network architecture is used here to predict the pm25_bin value, specifically using Pytorch's torch.nn package ("torch.nn"). The model consists of an input layer with two fully connected hidden layers with 32 and 16 nodes using ReLU activation functions. The output layer has 5 nodes representing each of the 5 bins identified for each sensor. A cross-entropy loss is used here as this is a classification problem but with ordinal classes. The node with the highest probability is selected and designated as the official PM 2.5 bin prediction for a given time point. This architecture is shared across all three sites, though the models are trained individually at each site with its given weather station.

Data from the overall time series is split into a train-test-validation setup. Extra precaution had to be taken given the time lag features that are used in the final model, as

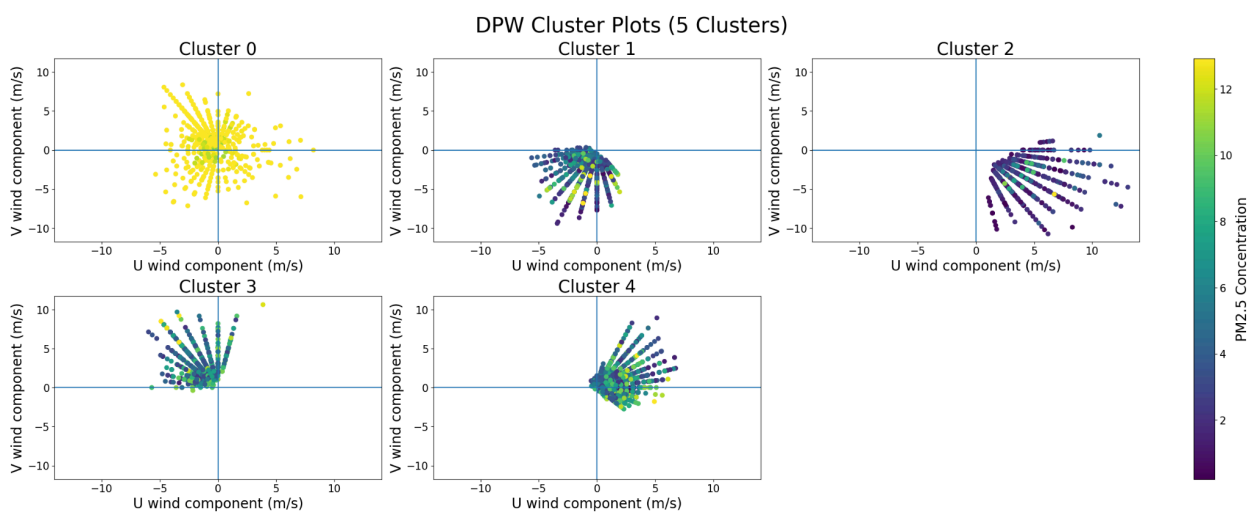
validation/testing data needs to be fully unseen to the training data. Originally, the split was made where the first 60% of the time series was training data, the following 20% was validation, and the final 20% was testing (with appropriate 7-hour gaps between to avoid data leakage). However, using warmer months as testing data and cooler months as training data led to skewed distributions of PM 2.5 data and thus skewed how many samples fell into each bin during testing. To obtain a more even distribution of PM 2.5 for training, testing, and validation data, the time series was split quarterly. Each of these quarters were then split into training, testing and validation splits and compiled together into the train/validation/test split used for the full model. This ensured that there was a more even distribution of features that may be correlated with seasonality.

Based on the validation and training accuracy, it was decided that each model would train for 1000 epochs with a learning rate of 0.001. As will be explained in the results section, the model did not initially perform well using only wind data. My goal then was to include other features such as the lags, relative humidity, and temperature as fit to make the model more accurate. The final model uses all the PM lag variables, wind_u and wind_v, hour_sin and hour_cos, and relative humidity to make its final prediction.

Results

Unsupervised Methods

The unsupervised clustering technique saw moderate to low success at identifying unique and informative wind-pollutant clusters. As seen in figure 2, at all three sites, the clustering was mainly based on wind direction with one cluster containing higher concentration measurements. Notably, the higher concentration cluster does not have a unique general wind direction associated with it. Clusters that are based around a single wind direction contained wider spreads of concentrations as well. However, it does seem that for all sites, wind coming from the northwest seems to be associated with lower PM 2.5 concentrations, though some minimal overlap with higher mean PM 2.5 concentrations exists.



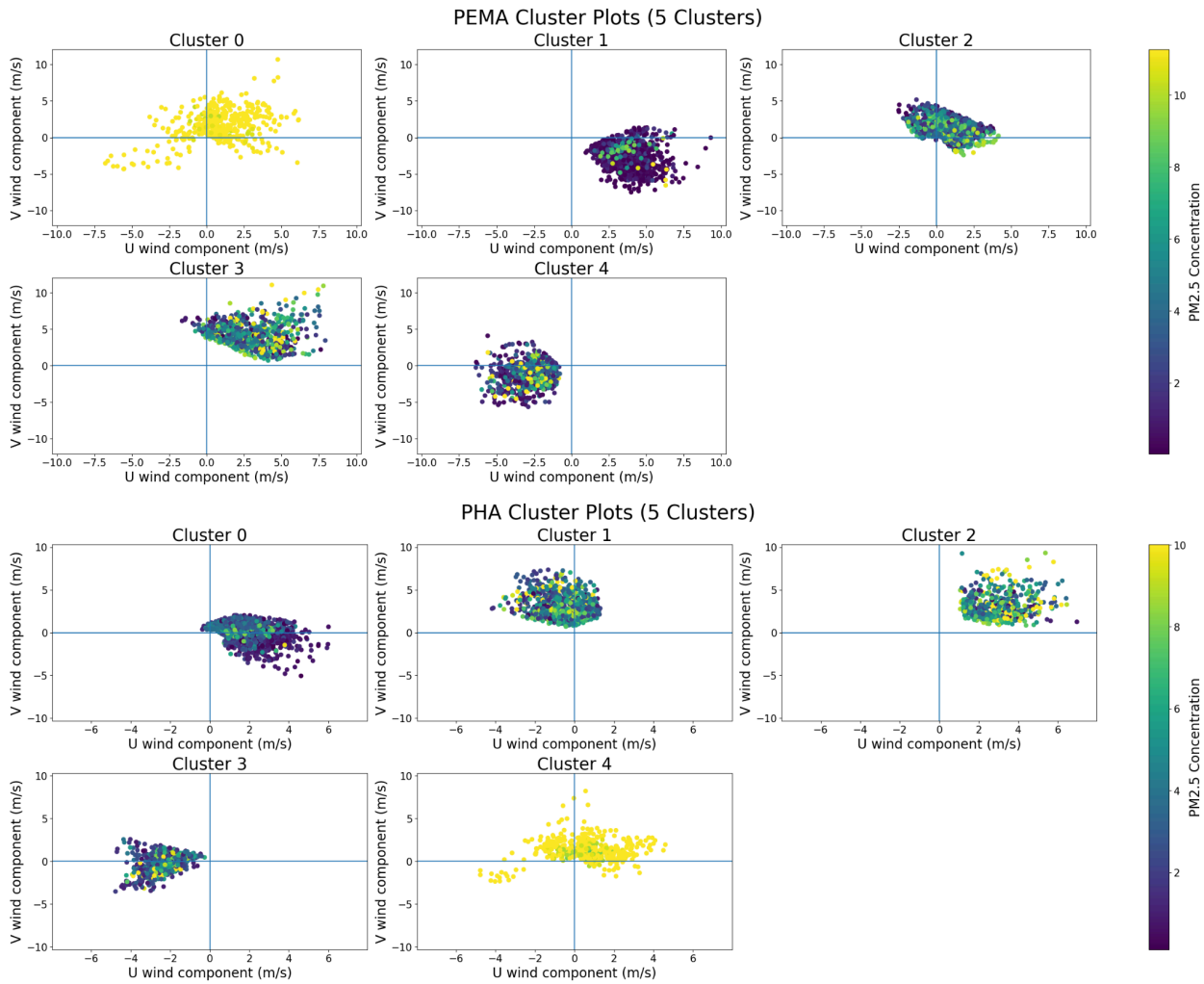


Figure 2: Cluster plots for PM 2.5. Each plot represents an identified cluster, and brighter colors correspond with higher PM 2.5 concentrations.

Experiments were conducted with 9 clusters, with additional experiments where humidity was included. These yielded similar results that were more based on wind direction and extreme concentrations. The large overlap in wind directions in the higher concentrations indicate that it would be difficult for a simple model to be able to predict concentration based on wind alone, which does seem to be true in the supervised learning section.

There is no baseline to compare this to, as it is meant to be informative, but this analysis could be expanded further in the future by considering more features in the clustering. Statistical tests can also potentially be used in these clusters to see if the wind direction and concentrations between the clusters are truly unique, or if there is too much overlap present.

Supervised Methods

The simple neural network with the parameters described in the methods section performed well with the sites on average having an accuracy of 75.43%. Figure 3 shows the confusion matrix for the three sites and table 3 in the appendix shows the summary statistics (precision, recall, F1 score) of each site.

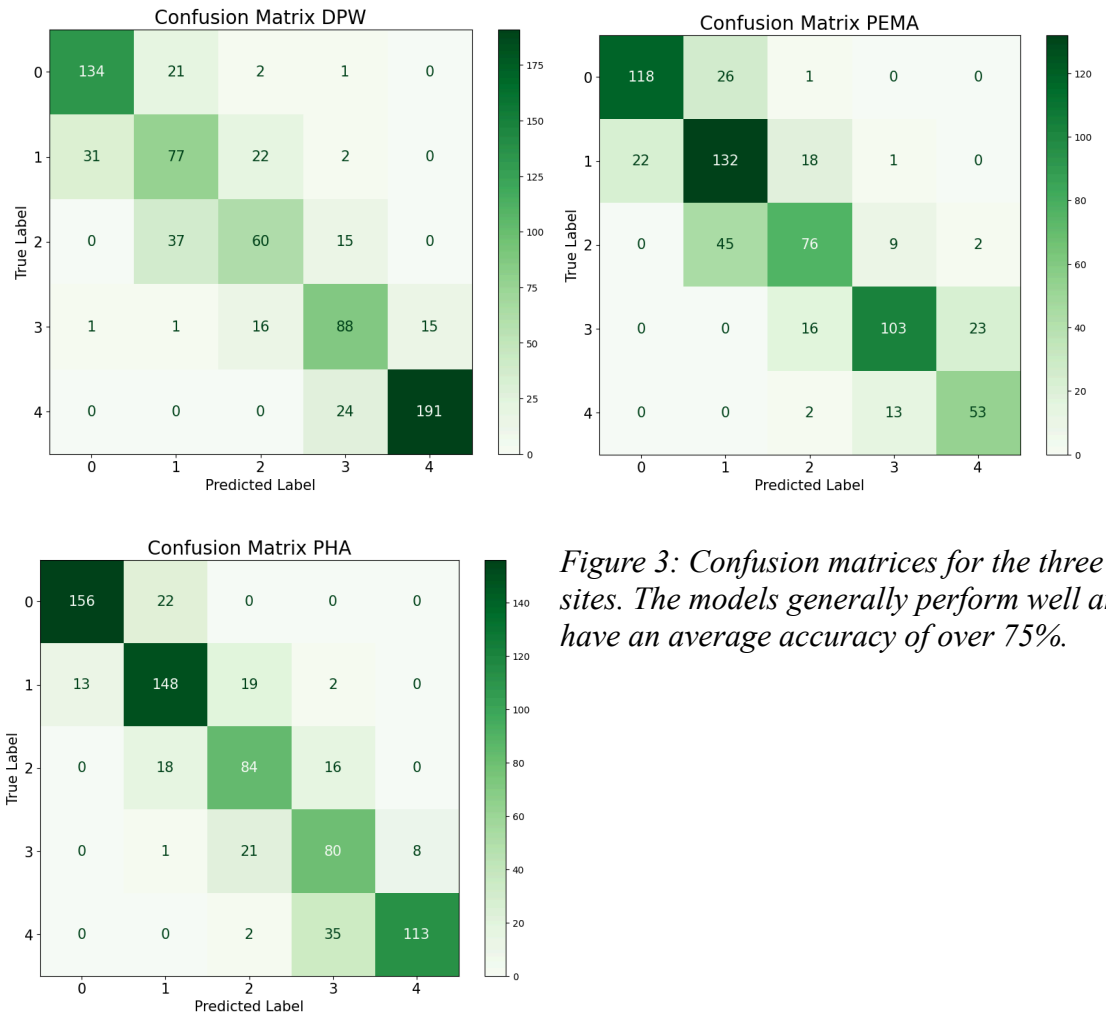


Figure 3: Confusion matrices for the three main sites. The models generally perform well and have an average accuracy of over 75%.

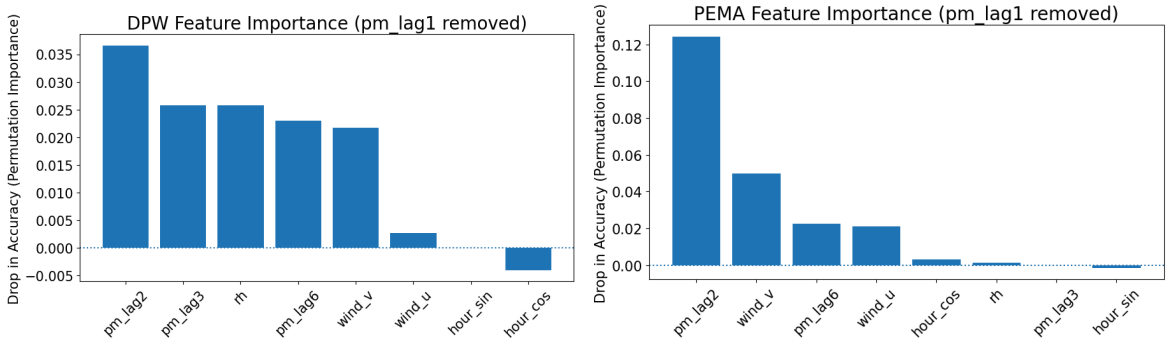
We can see that the model overall does a good job of predicting the lower bins and higher bins, with higher F1 scores relative to the middle concentration bins. Unfortunately, likely no baseline exists in literature to compare our results with. Almost all literature on PM 2.5 prediction, including those listed in the introduction, do not predict PM 2.5 into bins, rather they predict the continuous PM 2.5 concentration itself. Thus the standard is to use the coefficient of determination (R^2) and mean error statistics between predicted and actual values.

While a baseline may not exist in literature, we can develop one by considering our PM 2.5 bin prediction to simply be the bin PM 2.5 was in one hour prior. Thus, a pm_lag1_bin feature was developed and was used as the prediction. For all sites, this PM 2.5 baseline slightly outperformed the supervised models themselves, indicating a strong reliance on the one-hour PM 2.5 lag and generally indicating wind and other factors have little impact, and the PM 2.5 concentration is generally autocorrelated.

However, recall our goal is not necessarily to create a strong prediction model, but rather to understand the connection between wind and PM 2.5 regimes. Thus feature importance will still be considered here.

When the model was run with just wind values, the model performed rather poorly and would default to guessing a single bin for the vast majority of the data. Based on the clustering

techniques, I had assumed this would happen, and thus I intended to use XAI techniques to determine which features were the most impactful in making the PM predictions. There does exist literature that explores the predictive power of these features. As mentioned previously, Danesh Yazdi et al. utilize Shapley values to show that wind is one of the most important predictors of PM 2.5 concentration. I aim to show a similar result using a similar tactic, permutation importance.



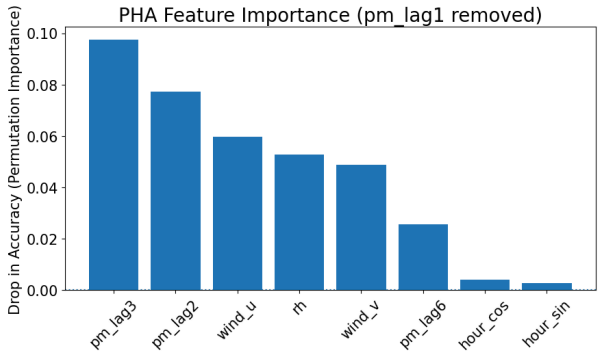


Figure 4: Feature importance at all three sites, with pm_lag1 removed. At two of the three sites, wind vector components have the most predictive power outside of PM 2.5 lags.

Permutation importance involves randomly permutating all data within an individual feature, as to ensure that the overall distribution of the feature is still the same, but the values are randomized. The difference in non-permutated accuracy and the permutated accuracy is the feature importance value. The pm_lag1 feature was the most important feature value by far, seeing about a 54.22% accuracy decrease when the features were randomized. This was expected based on EDA which shows that the pm_lag1 is strongly correlated with the actual PM readings, as seen in figure 5 in the appendix. Outside of the lags, wind vector components were among the most important features in the model for making predictions at two of the three sites. This higher importance of wind is generally in line with work done by Jin et al. and Danesh Yazdi et al. (though granted not using the exact same XAI techniques).

Discussion

There were several challenges faced when working on this project. Data availability was an issue primarily due to the timeline of when the Quants were set up in the field, as well as the outage of the Allens Ave NOAA meteorological site. This also made it more difficult to create a proper train-validation-test split on the data, and if this project were to be revisited it would be nice to use more data if available. There is also not as much documentation on the accuracy of the wind sensors on the Gill weather station that feeds data to the Quants.

In general, I believe this project can and will be expanded in several ways. For the sake of this project, I focused on using a neural network since it was an architecture I had not previously worked with for BP research. When only considering wind as the input to this model, it tended to predict one of the three middle bins (bins 1, 2, or 3) in order to minimize the error. I think it would be interesting to try more advanced off-the-shelf ML techniques like RandomForest or XGBoost to predict the PM concentration directly ("scikit-learn", "XGBoost"). This is beneficial as the prediction can still be fitted to PM bins for the sake of comparison to this project (by predicting continuous numerical values, then fitting them into bins afterwards). In addition, predicting the actual PM values allows for comparison to existing literature which primarily use standards like the coefficient of determination. If the other models are more successful, it would also be worth trying to refine the model to rely on larger time lags, which could potentially make the model better as an actual tool for forecasting PM 2.5 concentrations. I can envision this expanding to other sites not immediately collocated with wind sensors, and to other pollutants that the sensors can measure as well.

This project could also be reframed based on existing literature on what is considered dangerous levels of PM 2.5 concentrations. As seen in the appendix (table 2), there are only slight differences between concentrations that define each bin. A more advanced model may be able to more accurately predict the specific time frames when PM 2.5 is in a dangerous range based on wind conditions.

This project, while not a complete success, has shown that wind may still a somewhat important predictor for PM 2.5 in the city of Providence. The supervised model, while having high accuracy, depends heavily on knowing the PM 2.5 concentration from one hour before the prediction is made. In that sense, the model cannot serve to forecast PM 2.5 concentrations into the future, which was the original vision of the project. This project on its own cannot explore the "why" of PM 2.5 concentrations either, that is where and how are sources of PM 2.5 contributing to the overall concentrations. However, it was still gleaned that the wind components have high predictive power relative to other components of the model.

Data Availability

Full code, data, and additional analysis and visualizations are available at https://github.com/jacklardner/wind-pm-analysis. The QuantAQ map application is available here.

Appendix

Feature	Units	Description
rh	Percentage	Relative humidity. This is measured inside the sensor box and not through the Gill weather station. The humidity is relatively accurate to the outside humidity.
temp	Degrees Celcius	Temperature. This is measured inside the sensor box and not through the Gill weather station. Temperature is much warmer inside the box in warmer months and does not accurately describe outside weather conditions.
wind_u	Meters per second	U-component of measured wind speed. Used in unsupervised and supervised analysis.
wind_v	Meters per second	V-component of measured wind speed. Used in unsupervised and supervised analysis
hour_sin	Float value	Sine float presentation of the hour the observation occurs, meant to capture daily cycles.
hour_cos	Float value	Cosine float presentation of the hour the observation occurs, meant to capture daily cycles.
pm_lag1	Micrograms per meter cubed	PM 2.5 concentration measured one hour prior.
pm_lag2	Micrograms per meter cubed	PM 2.5 concentration measured two hours prior.
pm_lag3	Micrograms per meter cubed	PM 2.5 concentration measured three hours prior.
pm_lag6	Micrograms per meter cubed	PM 2.5 concentration measured six hours prior.
pm25	Micrograms per meter cubed	PM 2.5 concentration measured at a given time.

Table 1: Features used as input for supervised and unsupervised models. Not all features made it into the final supervised model, as discussed later. It should also be noted that the neural network was used to predict the PM 2.5 bins, not the concentration itself. The pm25 parameter above was used as input to the unsupervised method.

Site	Bin 0 Range (µg/m^3)	Bin 1 Range (µg/m^3)	Bin 2 Range (µg/m^3)	Bin 3 Range (µg/m^3)	Bin 4 Range (µg/m^3)
DPW	[0.22 - 2.29]	(2.29 - 3.68]	(3.68 - 5.64]	(5.64 - 9.39]	(9.39 - 66.98]
PEMA	[0.03 - 0.58]	(0.58 - 1.85]	(1.85 - 4.43]	(4.43 - 8.15]	(8.15 - 91.84]
PHA	[0.12 - 1.25]	(1.25 - 2.49]	(2.49 - 4.27]	(4.27 - 7.35]	(7.35 - 128.02]

Table 2: Range of values that define bins at each site. Quintiles are formed from the overall distribution and define the bins, with 0 being the lowest concentration and 4 being the highest concentration bin.

DPW				PEMA				PHA			
	Precision	Recall	F1 Score		Precision	Recall	F1 Score		Precision	Recall	F1 Score
Bin 0	0.81	0.85	0.83	Bin 0	0.84	0.81	0.83	Bin 0	0.92	0.88	0.90
Bin 1	0.57	0.58	0.57	Bin 1	0.65	0.76	0.70	Bin 1	0.78	0.81	0.80
Bin 2	0.60	0.54	0.57	Bin 2	0.67	0.58	0.62	Bin 2	0.67	0.71	0.69
Bin 3	0.68	0.73	0.70	Bin 3	0.82	0.73	0.77	Bin 3	0.60	0.73	0.66
Bin 4	0.93	0.89	0.91	Bin 4	0.68	0.78	0.73	Bin 4	0.93	0.75	0.83
Accuracy:	0.75			Accuracy:	0.73			Accuracy:	0.79		

Table 3: Classification report for three sites. The values here were produced from the Scikit Learn classification report package ("classification_report").

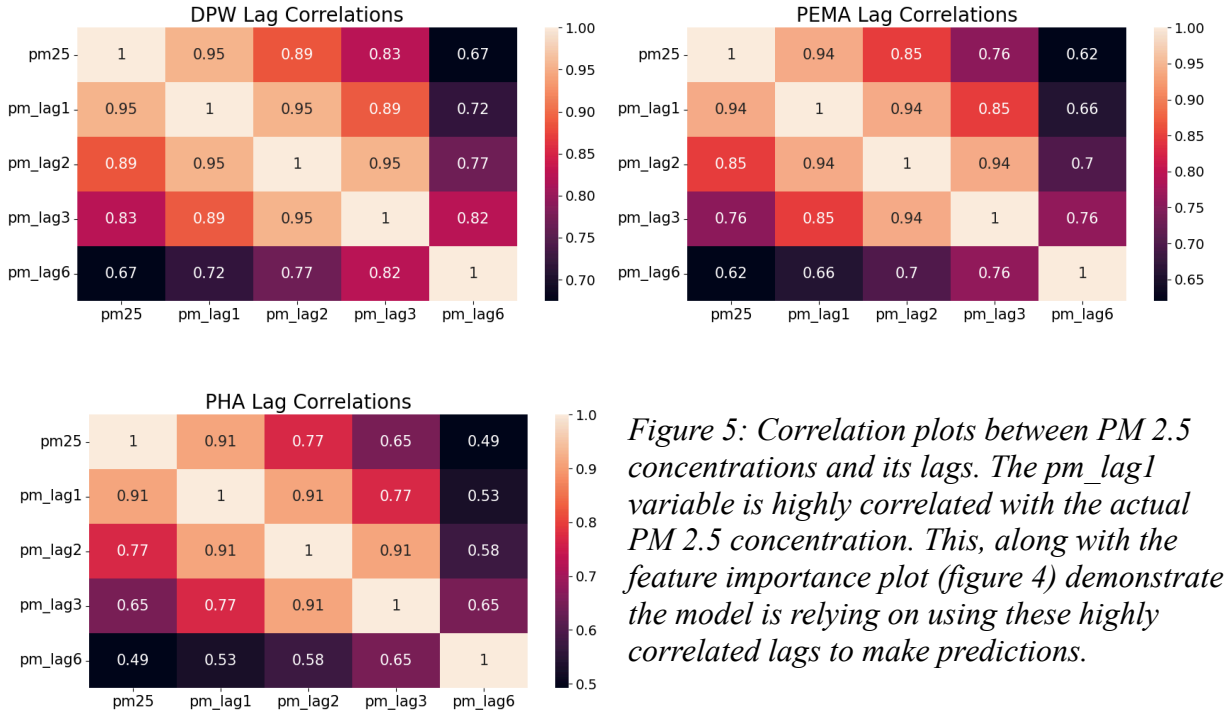


Figure 5: Correlation plots between PM 2.5 concentrations and its lags. The pm_lag1 variable is highly correlated with the actual PM 2.5 concentration. This, along with the feature importance plot (figure 4) demonstrate the model is relying on using these highly correlated lags to make predictions.

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